Claim Detection and Span Extraction in Reddit Health Discussions

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Abstract

Online health discussions frequently contain medical claims that are implied through personal narrative and experiential language rather than explicitly stated. Detecting such claims poses significant challenges for traditional argument mining systems, which assume well-bounded, formal assertions. In this paper, we present a modular NLP pipeline for claim detection and span extraction specifically tailored to noisy Reddit posts concerning self-medication. Our system operates in three stages: document-level binary claim detection, explicit vs. implicit claim typology classification, and token-level span extraction via a BIO tagging scheme. Initial baseline testing revealed that traditional machine learning models and BiLSTMs struggle severely with implicit claims, motivating our adoption of a contextual *BERT-mini* architecture. We evaluate our pipeline both in-domain on a curated Reddit corpus and zero-shot on the formal IBM Debater claim benchmark. Our best in-domain configuration achieves an F1 of 0.860, while the zero-shot cross-domain evaluation exposes a significant domain gap (F1: 0.326) and reveals a "Confidence Paradox," wherein the model exhibits higher confidence in its false predictions (0.469) than its correct ones (0.417). We supplement these findings with an input granularity ablation study and a qualitative error analysis, demonstrating the linguistic divergence between social media health claims and formal argumentation.

1 Introduction

Reddit and similar online health communities have become vast repositories of self-medication experiences, where users share anecdotal evidence, recommend off-label drug uses, and describe treatments in intensely personal terms. In the context of this study, a self-medication claim describes a user taking medication without medical guidance, changing their dosage, stopping prescribed medications, or mixing substances. Because unverified claims in these communities can lead to serious health risks, automatically identifying them is a pressing public health concern. Unlike formal argumentative text (e.g., Wikipedia articles or legal documents), social media health claims are highly contextual, noisy, and often embedded within long personal narratives. They rely heavily on implicit formulation (e.g., *"St. John's Wort saved my life"*) rather than direct, formal assertions.

Traditional argument mining models often struggle with this domain shift, as they are trained to identify explicit evaluative language and strict logical structures. To address this gap, we propose a modular, three-stage NLP pipeline designed to extract and classify medical claims from unstructured social media discourse.

Before adopting a transformer-based approach, we established performance baselines using FastText embeddings (`crawl-300d-2M-subword`) paired with Logistic Regression, Naive Bayes, and a BiLSTM network. While these models performed adequately on explicitly formulated statements, their performance degraded drastically on implicit claims. This stark discrepancy underscored the necessity of deep contextual representations to capture pragmatic intent, directly motivating our shift to a *BERT-mini* pipeline.

Our approach separates the task into three distinct sub-problems: (1) **Binary Claim Detection** to filter irrelevant posts, (2) **Claim Typology Classification** to categorize claims as explicit or implicit, and (3) **Token-Level Span Extraction** to isolate the exact linguistic boundaries of the claim. By formulating span extraction as a token classification task using a BIO (Begin, Inside, Outside) tagging scheme, our pipeline moves beyond document-level categorization to provide interpretable, granular insights into claim structure.

We further evaluate the generalizability of our

Reddit-trained model by testing it zero-shot against the IBM Debater dataset, a high-quality corpus of formal argumentative claims. This cross-domain evaluation yields our most significant finding: the **Confidence Paradox**. While the model successfully learns the linguistic signatures of Reddit health claims, it fails to generalize to formal arguments, becoming miscalibrated and overly confident in its erroneous predictions on out-of-domain text.

In summary, the primary contributions of this paper are:

- A three-stage transformer-based pipeline (using *BERT-mini*) capable of detecting, typing, and extracting medical claims from noisy Reddit discourse.
- An input granularity ablation study demonstrating the trade-offs between precision-oriented and recall-oriented inference strategies.
- A claim typology analysis showing that claims are linguistically heterogeneous, falling into Causal, Contrastive, Epistemic, Normative, and Multi-label categories.
- A zero-shot cross-domain evaluation highlighting the Confidence Paradox and quantifying the domain gap between social media self-medication claims and formal argumentation.
- A hedging analysis examining the prevalence and distribution of epistemic uncertainty markers across explicit and implicit claim spans.

2 Related Work

Our research focuses between two primary subfields of natural language processing, argument mining and health discourse analysis on social media.

2.1 Argument Mining and Claim Detection

Argument unit identification is a foundational task in argument mining, traditionally focused on formal, well-structured text such as legal documents, persuasive essays, or Wikipedia articles. Classic approaches aim to uncover the structure of arguments by identifying claims and supporting premises, relying on supervised, task-specific classifiers trained on domain-specific corpora. Recent work has examined how large language models reshape this landscape shifting from pipeline-based classifiers toward prompt-driven and reasoning-oriented paradigms while also noting that task boundaries blur significantly when models are applied to informal or out-of-domain text (Li et al., 2025). Applying these formal argumentation frameworks to user-generated web content remains challenging due to the absence of strict logical structures and the prevalence of contextual, implicit reasoning.

2.2 Health Discourse and Self-Disclosure on Social Media

Recent works are focused towards mining of health-related claims and self-disclosures from social media platforms. Wührl and Klinger (2021) annotated a corpus of biomedical tweets for implicit and explicit claims, demonstrating the difficulty of applying standard transfer learning approaches such as BERT to noisy biomedical text. Similarly, Romano et al. (2023) highlighted the utility of extracting clinically relevant keyphrases from self-reported health experiences online.

Specific to the Reddit domain, Yang et al. (2024) developed models to identify phases of opioid use and addiction through self-disclosures, emphasizing the importance of span-level extractive explanations for interpretability. Mikaelsen and Åsheim (2024) explored the detection of self-diagnosis and self-medication behaviors about ADHD on Reddit and TikTok, showing the domain-specific linguistic patterns that is seen in peer-support communities.

Our work bridges these threads. While prior studies have either classified medical posts at the document level or identified broad argumentative components in formal text, we propose a granular, three-stage pipeline that detects self-medication claims, categorizes their linguistic typology (explicit vs. implicit), and finds their textual spans combining the health discourse mining with the structural rigour of argument mining.

Study	Domain	Typology
Wührl and Klinger (2021)	Biomedical tweets	✓
Romano et al. (2023)	Reddit/social	✗
Yang et al. (2024)	Reddit (opioids)	✗
Mikaelsen and Åsheim (2024)	Reddit/TikTok	✗
This work	Reddit (self-med)	✓

Table 1: Comparison of related work by domain and claim typology classification. ✓ = supported, ✗ = not addressed.

3 Dataset and Task Definition

To train and evaluate our in-domain models, we used a dataset focused on self-medication and health discussions. Formulating our pipeline tasks required an annotation process tailored to the unique noise, colloquialisms, and implicit reasoning typical of social media discourse.

3.1 Data Collection and Source Justification

To construct our in-domain corpus, we collected an initial set of 5,088 Reddit posts using the PullPush API. To ensure a diverse representation of health discourse, we sourced data from nine subreddits spanning three distinct medical domains (the full list of subreddits and search terms is detailed in Appendix A and Figure 1):

1. **General Medical Advice:** Communities showing inquiries about over-the-counter interactions and self-managed symptom relief.

2. **Mental Health:** Forums where users discuss self-medicating for psychiatric symptoms, altering the doses prescribed to them, or changing medications due to side effects.

3. **Substance Recovery:** Communities containing dense, highly implicit narratives regarding tapering, withdrawal management, and off-label substance use.

To distill this raw collection into a high-density corpus, we applied a regex-based filtering strategy targeting three linguistic axes: Action Patterns (verbs denoting self-treatment), Substance Patterns (mentions of unprescribed drugs), and Target Conditions (specific ailments). This filtering reduced the initial collection to 1,215 highly relevant posts for manual annotation.

3.2 Dataset Description and Statistics

The final curated corpus (`labels_v5.csv`) exhibits a strong, realistic class imbalance typical of social media mining (see Table 2).

Within the 1,215 posts, genuine medical claims are a minority relative to conversational noise. There is also a severe skew toward explicit formulation: our annotation identified 597 explicit claim spans compared to only 53 implicit spans. This imbalance reflects the natural rarity of strictly implicit medical reasoning and makes it a particularly challenging target for machine learning classifiers.

Dataset Characteristic	Count
Total Posts Annotated	1,215
Total Claims Identified	650
Claim Typology Breakdown	
Explicit Claims	597
Implicit Claims	53

Table 2: Distribution of labels in the curated Reddit self-medication dataset. The severe imbalance between explicit and implicit claims highlights the rarity and difficulty of purely implicit reasoning.

3.3 Annotation Guidelines and Challenges

The annotation task was to identify statements regarding self-medication that shows effectiveness, risks, necessity, or justification. To distinguish genuine claims from surrounding noise, the guidelines enforced the following rules.

Self-Report vs. Self-Claim. A statement is labeled a *Self-Report* (Non-Claim) if it merely describes actions without evaluating outcomes. A *Self-Claim* infers causality or expresses a judgment about those outcomes.

> **Self-Report (Non-Claim):** *"I took ibuprofen twice a day."* (Pure narration, no outcome evaluated.)
>
> **Self-Claim (Claim):** *"Ibuprofen helped my headache."* (Direct evaluation of outcome.)

Claim Typology (Explicit vs. Implicit). Claims were categorized based on their linguistics. *Explicit Claims* shows direct evaluative language and clear verbs of judgment (e.g., *helped*, *effective*).

> **Explicit Claim:** *"Migraines and anxiety... Buvidal saved me and my career and I take it monthly to this day."* (Direct evaluative verb: *saved*.)

Implicit Claims lack direct evaluative verbs but allow effectiveness to be logically inferred from surrounding pragmatic context.

> **Implicit Claim:** *"The only way I could ever stop is the CT for three or four days and then going half ration. That was hard but doable."* (No explicit evaluative verb, but a successful outcome is logically inferred.)

Where a post contained both claim types, explicit claims were given annotation priority, as they provide less ambiguous ground truth for the span extraction task.

Annotation Challenges and Label Noise. Given the informal, multi-sentence nature of Reddit posts, the annotation process was susceptible to subjectivity. Two primary challenges emerged:

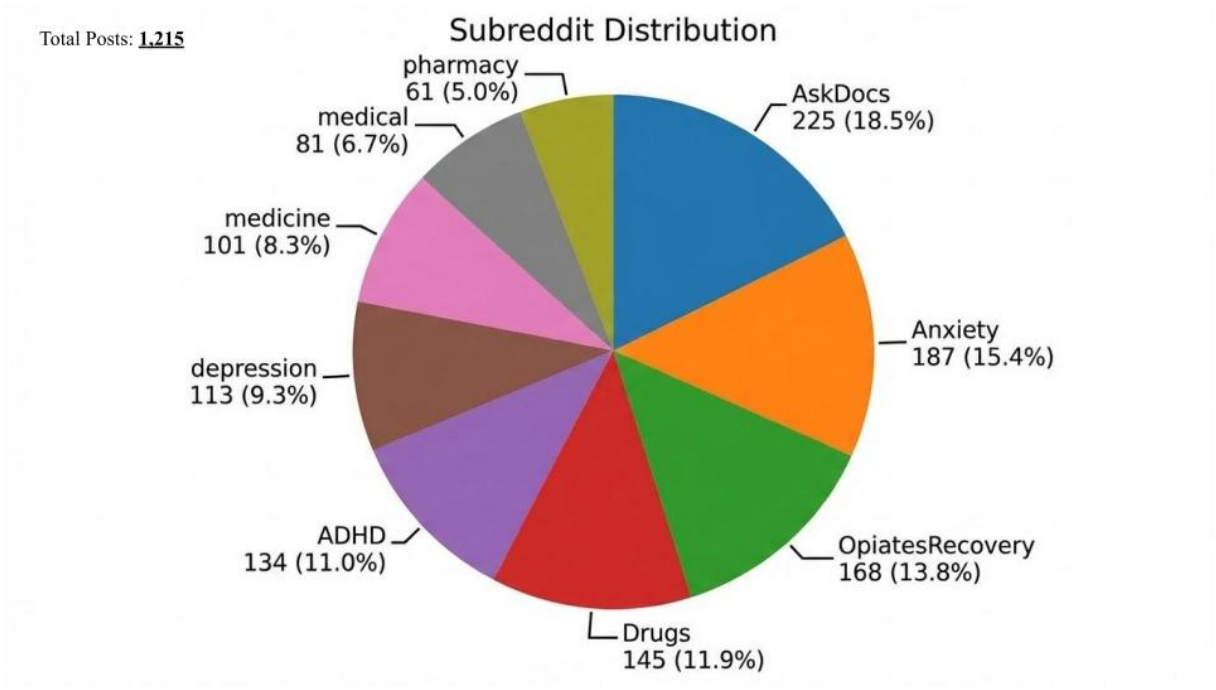

Figure 1: Distribution of r/subreddits extracted using the PullPush API.

1. **Subjectivity of Implicit Claims:** Determining the threshold at which a personal narrative transforms into an implicit claim relies heavily on the annotator's interpretation.

2. **Fuzzy Span Boundaries:** Claims made on social media often split between broken sentences. Annotators frequently disagreed on exact token boundaries, leading to "fuzzy" ground-truth spans.

3.4 Task Formalization

Based on the annotated dataset, three sub-tasks were defined that form our pipeline:

1. **Binary Claim Detection (Branch 1):** A document-level binary classification task to check whether a post contains at least one self-medication claim or not.

2. **Claim Typology Classification (Branch 2a):** Given a post flagged as containing a claim, classify its linguistic formulation as *Explicit* or *Implicit*.

3. **Span Extraction (Branch 2b):** A token-level classification task to extract the exact phrase constituting the claim, using the BIO tagging scheme: **B** marks the beginning token of a claim span, **I** marks continuation tokens, and **O** denotes non-claim tokens. This granular extraction also acts as a regularizer against the fuzzy boundary noise described above.

4 Methodology

To handle the noisy, informal, and highly implicit nature of self-medication claims on Reddit, we propose a cascading, three-stage inference pipeline. By decomposing the problem into distinct subtasks, we reduce error propagation and allow for specialized representations at each step.

All three branches utilize the lightweight *BERT-mini* architecture (`prajjwal1/bert-mini`) as their shared contextual encoder. This choice is motivated by two considerations. First, BERT fine-tuning consistently outperforms across a range of NLP benchmarks (Devlin et al., 2019). Second, *BERT-mini* offers substantially reduced parameter count relative to *BERT-base*, enabling fast inference at the scale required for large social media corpora, a practical necessity when processing thousands of Reddit posts.

4.1 Binary Claim Detection

The first branch acts as a document-level gatekeeper, filtering out the large volume of conversational noise, pure medical inquiries, and supportive narrations that dominate health subreddits. It is formulated as a sequence classification task: the *BERT-mini* encoder processes the full post text and produces an aggregated sequence representation

via the [CLS] token, which is passed through a linear classification head to output a binary probability (Claim vs. Non-Claim).

During inference, posts falling below a confidence threshold of 0.5 are discarded and the pipeline halts, conserving computation. Posts above the threshold are simultaneously routed to Branch 2a and Branch 2b.

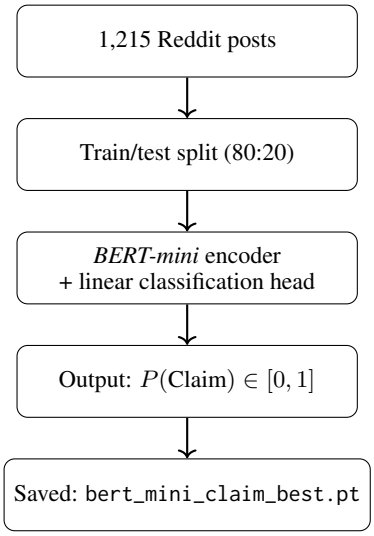

Figure 2: Branch 1: Binary Claim Detection pipeline.

4.2 Claim Typology Classification

Once a post is detected as containing a claim, Branch 2a then classifies it as *Explicit* or *Implicit*. This step distinguishes between direct medical evaluations and inferred outcomes.

Training a dedicated model for this task rather than a single multi-task classifier is a deliberate design choice. As noted in Section 3, implicit claims are exceedingly rare (53 of 650 annotated claims). Forcing a single model to learn binary claim detection alongside this extreme class imbalance frequently causes the minority class to be ignored entirely. By isolating typology in Branch 2a, the model can focus its representations on the subtle epistemic markers such as modal auxiliaries (*might*, *could*) and hedging phrases (*I think*) that differentiate a direct assertion from a narrative conclusion. Class-weighted loss is applied during training to further mitigate the imbalance.

4.3 Token-Level Span Extraction

The Branches 1 and 2a work at the document level, Branch 2b finds the exact boundaries of the detected claim within the post. This changes the problem as token-level classification using the BIO

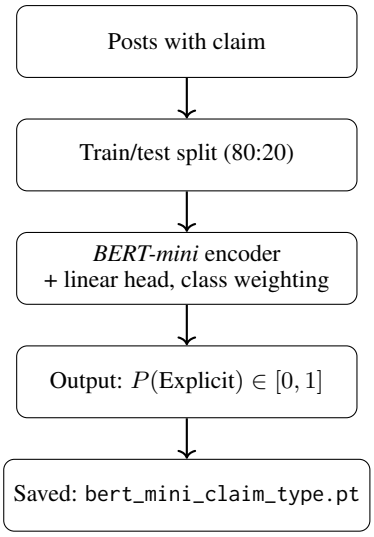

Figure 3: Branch 2a: Claim Typology Classification (Explicit vs. Implicit).

tagging scheme, a standard formulation for span extraction tasks such as named entity recognition (Tjong Kim Sang and De Meulder, 2003). A classification head assigns one of three labels to each token: **B** (claim begin), **I** (claim continuation), or **O** (non-claim).

There are two uses for this design. First, it resolves the fuzzy boundary problem described in Section 3 by requiring the model to make strict, token-by-token decisions rather than selecting from pre-defined span candidates. Second, enforcing precise span boundaries acts as an implicit regularizer during inference, reducing the tendency to hallucinate claim regions in non-claim text.

5 Results

We evaluate our pipeline on both Reddit data and an out-of-domain IBM benchmark. All *BERT-mini* models were fine-tuned with a learning rate of 2×10^{-5} and a confidence threshold of 0.5 at inference time.

5.1 Baseline Comparisons

To see is the contextual embeddings are necessary for this task, we compare *BERT-mini* against three simpler baselines using pre-trained FastText embeddings (crawl-300d-2M-subword) (Bojanowski et al., 2017): Logistic Regression, Gaussian Naive Bayes, and a Bi-directional LSTM (BiLSTM). All models were trained on the same dataset for both binary claim detection and claim typology classification.

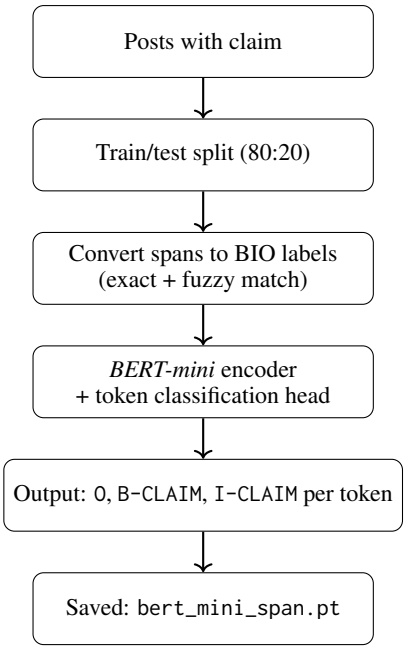

Figure 4: Branch 2b: Token-level Span Extraction using BIO tagging.

Model	Explicit F1	Implicit F1
Logistic Regression	0.78	0.21
Naive Bayes	0.87	0.13
BiLSTM	0.85	0.12
BERT-mini (ours)	**0.92**	**0.28**

Table 3: Per-class F1 scores on claim typology classification. All baselines use FastText embeddings; *BERT-mini* uses contextual representations.

As shown in Table 3, all models perform reasonably well on explicit claims (F1 $\geq$ 0.75), where surface-level lexical cues such as evaluative verbs are sufficient signals. Performance drops sharply for implicit claims across all baselines: Naive Bayes and BiLSTM reach F1 scores of only 0.13 and 0.12 respectively, confirming that static embeddings and recurrent architectures cannot capture the pragmatic, cross-sentence context on which implicit claims depend.

BERT-mini improves implicit claim F1 to 0.28 while maintaining strong explicit performance (0.92), demonstrating that contextual representations capture discourse signals unavailable to static models. The persistently low implicit F1 across all models suggests that the task likely requires deeper pragmatic reasoning beyond what sentence-level fine-tuning provides a limitation we revisit in Section 7.

5.2 In-Domain Branch Performance

Table 4 reports the standalone performance of each pipeline branch on the curated Reddit corpus.

Branch	Task	F1
Branch 1	Claim vs. No-Claim (Macro F1)	0.85
Branch 2a	Explicit vs. Implicit	0.56
Branch 2b	Span extraction	0.82

Table 4: Standalone branch performance on the Reddit self-medication dataset.

Branch 1 achieves a Macro F1 of 0.85, indicating that the document-level claim detector generalizes well to the informal register of Reddit despite never seeing formal argument structure. Branch 2b reaches F1 of 0.82 on span extraction, suggesting that BIO tagging effectively constrains the model's predictions even under fuzzy ground-truth boundaries.

Branch 2a is the weakest component at F1 of 0.56. This is expected: distinguishing explicit from implicit claims requires sensitivity to subtle epistemic markers and cross-sentence inference that a sentence-level classifier handles poorly, particularly given the severe class imbalance (597 explicit vs. 53 implicit spans in the training data). The gap between Branch 2a and the other branches points to typology classification as the primary bottleneck of the pipeline.

To illustrate the end-to-end pipeline behaviour, we present three constructed examples below. The drug name *ozempdol* in Example 3 is fictional and used solely for illustration.

Example 1: Pure Narration (No Claim)

"I study in Potsdam, I take medicine for anxiety."

Output: `{claim: False, reason: 'no reliable claim sentence detected'}`

Example 2: Contextual Inference (Implicit Claim)

"I study in Potsdam, I take medicine for anxiety. Before that, I couldn't leave my apartment for weeks. My exams went much better this semester."

Output: `{claim: True, confidence: 0.716, type: 'implicit', span: "Before that, I couldn't leave my apartment for weeks. My exams went much better this semester."}`

Example 3: Direct Assertion (Explicit Claim)

"I study in Potsdam... My exams went much better this semester. So taking ozempdol cures anxiety

but if taken in small doses."

Output: `{claim: True, confidence: 0.661, type: 'explicit', span: "So taking ozempdol cures anxiety but if taken in small doses."}`

6 Experiments and Analysis

We conducted both in-domain and zero-shot cross-domain experiments to evaluate the pipeline's performance and generalizability.

6.1 In-Domain Performance and Ablation Analysis

Because user-generated posts vary drastically in length and distribute context unevenly across disjointed thoughts, determining the optimal input granularity for inference is non-trivial. We conducted an input granularity ablation study testing three distinct inference variants (Figure 5):

- **Variant 1 - Full Post:** The entire Reddit post is processed as a single contiguous unit. The post is labeled as containing a claim if the document-level classifier exceeds the confidence threshold.
- **Variant 2 - Any Sentence:** The post is split into individual sentences using regular expressions. Each sentence is classified independently, and the post is flagged if *any* sentence exceeds the confidence threshold. This is a recall-oriented strategy.
- **Variant 3 - Best Sentence:** Sentences are ranked by classifier confidence and the highest-scoring sentence is selected as the representative span. Additional constraints on span length are enforced, making this a precision-oriented strategy.

Inference Strategy	Precision	Recall	F1
Full Post	0.801	**0.928**	**0.860**
Any Sentence	0.831	0.740	0.783
Best Sentence	**0.868**	0.263	0.404

Table 5: Input granularity ablation on the Reddit dataset.

As shown in Table 5, the three variants expose a clear precision–recall trade-off governed by how much context the classifier receives. The Full Post variant achieves the strongest overall F1 (0.860) by preserving cross-sentence context, which is critical for implicit claims. An isolated sentence such as *"I stopped taking the prescribed dosage"* reads as pure narration, but paired with *"It was the only way I could get out of bed"* it forms an unambiguous implicit claim. Fragmenting the post strips away these pragmatic clues, explaining the recall drop in both sentence-level variants.

The Any Sentence variant trades recall for a modest precision gain, while the Best Sentence variant pushes this further: by demanding highly concentrated evidence within a single span, it reaches the highest precision (0.868) but at the cost of catastrophic recall (0.263), yielding an F1 of only 0.404. This confirms that Best Sentence functions as an aggressive regularizer suitable only when false positives are especially costly and missing contextualized claims is acceptable.

6.2 Claim Typology (In-Domain)

Not all claims are linguistically identical. Applying our claim typology heuristics to the annotated Reddit spans, we observed the following distribution across four categories:

- **Causal (195) and Contrastive (204):** Together these dominate the corpus, reflecting the health-reasoning patterns typical of self-medication discourse (e.g., *"X leads to Y"* or *"Just because X doesn't mean Y"*).
- **Epistemic (110):** Uncertainty and belief markers (e.g., *"I think," "might"*) are common, which directly underlies the difficulty of implicit claim detection.
- **Normative (15):** Prescriptive statements (*"You should..."*) are rare, indicating that Reddit users share personal reasoning far more frequently than direct medical advice.
- **Multi-label (126):** Approximately 18% of claims combine causal, contrastive, and epistemic markers simultaneously, demonstrating that strictly binary typology labels underrepresent the pragmatic richness of the data.

We assign typology labels using the rule-based surface patterns detailed in Appendix D.

6.3 Zero-Shot Cross-Domain Evaluation

To assess whether the pipeline has learned general argumentation patterns or merely Reddit-specific surface features, we evaluate the Reddit-trained Branch 1 classifier zero-shot on the IBM Debater

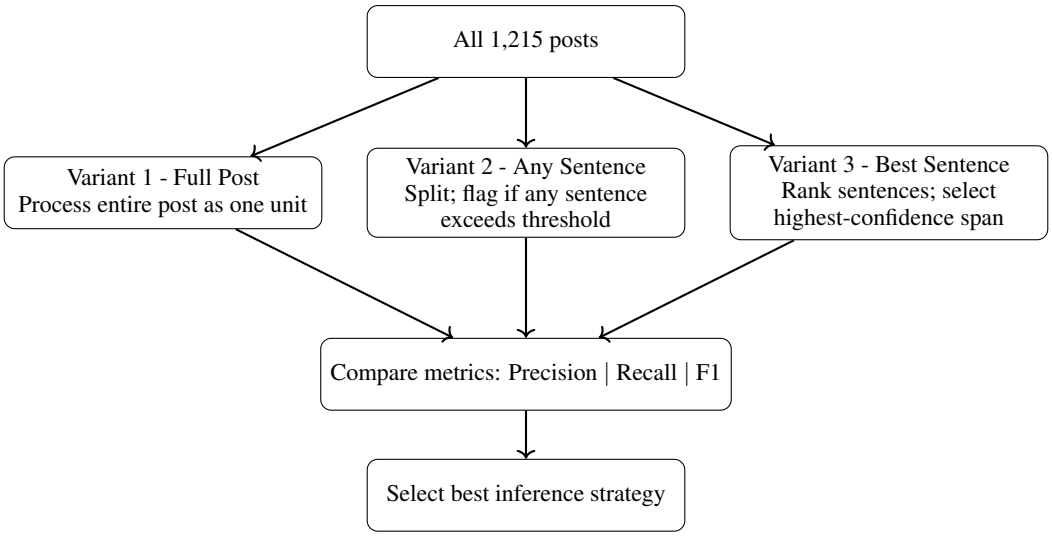

Figure 5: The three variants used in the input granularity ablation study.

dataset without any additional fine-tuning. Zero-shot cross-domain evaluation is a standard diagnostic for generalization, as it surfaces domain-specific overfitting that in-domain tests cannot detect (Ramponi and Plank, 2020).

6.3.1 The Domain Gap: Social Media vs. Wikipedia

The IBM Debater dataset consists of formal argumentative claims extracted from Wikipedia articles, presenting a sharp linguistic contrast to our Reddit corpus. Reddit users rely on first-person narratives, experiential evidence, and implicit causality, whereas Wikipedia claims are characterized by third-person objectivity, institutional reporting, and explicit logical assertions.

Metric	Zero-Shot Score
Accuracy	0.654
Precision	0.380
Recall	0.285
F1	0.326

Table 6: Zero-shot performance of the Reddit-trained pipeline on the IBM Debater dataset.

As shown in Table 6, the model achieves an F1 of 0.326, a sharp drop from the in-domain F1 of 0.860 (Full Post variant). The low recall (0.285) indicates the model is largely blind to formal claims, which lack the personal evaluative markers it was trained on. The low precision (0.380) indicates that formal medical vocabulary terms such as *genes*, *alcoholism*, and *treatment* triggers false positive claim predictions even when the surrounding sentence is purely factual.

6.3.2 The Confidence Paradox in Out-of-Domain Generalization

A good model should show a higher confidence to correct predictions than to incorrect ones. Our evaluation logs reveal the opposite behavior when the Reddit-trained classifier processes IBM Debater text, a phenomenon we term the **Confidence Paradox**.

Extracting the sigmoid output logits from Branch 1, we find:

- Average confidence when **correct**: 0.417
- Average confidence when **wrong**: 0.469

The model is systematically more confident in its errors than in its correct predictions. This miscalibration arises because the classifier has overfit to shallow lexical cues specific medical keywords that co-occur with claims in Reddit data rather than learning structural features of argumentation. When those keywords appear in a formal Wikipedia sentence with factual or speculative framing, the model fires confidently despite the surrounding context contradicting a claim reading.

6.4 Hedging Analysis

Hedging detection has a well-established precedent in biomedical NLP, with epistemic markers such as modal auxiliaries and evidential verbs serving as primary indicators of speculative assertions in health corpora (Szarvas et al., 2008). We analyzed hedging prevalence across both explicit and implicit claim spans to determine whether epistemic uncertainty is a typology-specific signal.

Of the 597 explicit spans, 114 (19.1%) contained at least one hedging marker, with a mean of 0.23 hedges per span. Of the 53 implicit spans, 10 (18.9%) contained hedging, with a mean of 0.25 hedges per span. The near-identical rates across typologies indicate that epistemic uncertainty is a domain-wide property of online health discourse rather than a reliable marker of implicit formulation. Consequently, systems that use hedging language as a proxy for claim typology will likely underperform on this corpus, as the signal does not discriminate between the two classes.

7 Limitations and Ethical Considerations

7.1 Limitations

Annotation Subjectivity and Fuzzy Boundaries. As discussed in Section 3, determining where a personal narrative becomes an implicit claim relies heavily on annotator interpretation. This subjectivity extends to span boundaries: annotators frequently disagreed on whether to include preceding context clauses, meaning BIO evaluation metrics may under-represent the model's true communicative understanding when a predicted span is pragmatically correct but token-misaligned with the ground truth.

Dataset Scope and Filtering Bias. The corpus is limited to English posts from nine Reddit communities, and the regex-based filtering strategy biases the data toward recognizable self-medication behaviors. Users describing off-label substance use through niche slang or purely implicit narratives without standard keywords are likely underrepresented.

Cross-Domain Generalization. As demonstrated in Section 6.3, *BERT-mini* overfits to social media stylistic conventions (first-person pronouns, specific medical vocabulary), failing to generalize to formal argumentative structures. This manifests directly as the Confidence Paradox.

7.2 Ethical Considerations

Privacy and Anonymization. Although data was collected from public Reddit forums via the PullPush API, we removed all URLs, Markdown links, and user mentions (`u/username`) from the corpus to prevent de-anonymization of individuals disclosing sensitive health information.

Intended Use and Deployment Risks. This pipeline is intended solely as an observational tool for public health researchers and computational linguists. It is **not** designed for medical diagnosis, claim verification, or automated content moderation. Deploying it for moderation could inadvertently silence vulnerable users seeking help or sharing recovery narratives.

8 Conclusion

We presented a lightweight, three-stage NLP pipeline for detecting, typing, and extracting self-medication claims from Reddit health discussions using a shared *BERT-mini* encoder. Our in-domain ablation demonstrates that claim detection relies critically on cross-sentence context: the Full Post variant achieves F1 of 0.860, while sentence-level fragmentation degrades recall substantially. Token-level BIO span extraction further constrains predictions and reduces hallucination under fuzzy ground-truth boundaries.

Zero-shot evaluation on the IBM Debater dataset exposes a significant generalization barrier (F1: 0.326), revealing that the model overfits to Reddit-specific lexical cues rather than learning universal argument structure. This manifests as the Confidence Paradox: the model assigns higher average confidence to its errors (0.469) than to its correct predictions (0.417) on out-of-domain text.

These findings point to a broader challenge for computational argumentation: detecting medical claims in informal discourse requires pragmatic reasoning over context, not surface-level keyword matching. Implicit claims remain the hardest target, and closing the gap between social media and formal argumentation domains is a necessary step toward robust, deployable claim detection systems.

References

Piotr Bojanowski, Edouard Grave, Armand Joulin, and Tomas Mikolov. 2017. Enriching word vectors with subword information. *Transactions of the Association for Computational Linguistics*, 5:135–146.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.

Hao Li, Viktor Schlegel, Yizheng Sun, Riza Batista-Navarro, and Goran Nenadic. 2025. Large language

models in argument mining: A survey. *Preprint*, arXiv:2506.16383.

Hilde Mikaelsen and Vår Åsheim. 2024. Dr. Social Media: Detecting self-diagnosis and self-medication of ADHD on social media using natural language processing. Master's thesis, Norwegian University of Science and Technology (NTNU).

Alan Ramponi and Barbara Plank. 2020. Neural unsupervised domain adaptation in NLP—A survey. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 6838–6855, Barcelona, Spain (Online). International Committee on Computational Linguistics.

William Romano, Omar Sharif, Madhusudan Basak, Joseph Gatto, and Sarah Preum. 2023. Theme-driven keyphrase extraction to analyze social media discourse. *Preprint*, arXiv:2301.11508.

György Szarvas, Veronika Vincze, Richárd Farkas, and János Csirik. 2008. The BioScope corpus: Annotation for negation, uncertainty and their scope in biomedical texts. In *Proceedings of the Workshop on Current Trends in Biomedical Natural Language Processing*, pages 38–45, Columbus, Ohio. Association for Computational Linguistics.

Erik F. Tjong Kim Sang and Fien De Meulder. 2003. Introduction to the CoNLL-2003 shared task: Language-independent named entity recognition. In *Proceedings of the Seventh Conference on Natural Language Learning at HLT-NAACL 2003*, pages 142–147.

Amelie Wührl and Roman Klinger. 2021. Claim detection in biomedical Twitter posts. In *Proceedings of the 20th Workshop on Biomedical Language Processing*, pages 131–142. Association for Computational Linguistics.

Chenghao Yang, Tuhin Chakrabarty, Karli R Hochstatter, Melissa N Slavin, Nabila El-Bassel, and Smaranda Muresan. 2024. Identifying self-disclosures of use, misuse and addiction in community-based social media posts. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 2507–2521. Association for Computational Linguistics.

A Data Collection and Filtering Details

A.1 Source Subreddits

Data was collected from the following nine Reddit communities to ensure a diverse representation of health and self-medication discourse: *r/AskDocs*, *r/Anxiety*, *r/ADHD*, *r/depression*, *r/Drugs*, *r/OpiatesRecovery*, *r/medicine*, *r/medical*, and *r/pharmacy*.

A.2 API Collection Keywords

The initial dataset of 5,088 posts was gathered using the PullPush API with the following search terms: *"self medication"*, *"self medicating"*, *"self-medication"*, *"OTC"*, *"over the counter"*, *"over-the-counter"*, *"self treatment"*, *"self-treatment"*, *"home remedies"*, *"self care"*, *"self-care"*, *"no prescription"*, *"without prescription"*, and *"took it without doctor"*.

A.3 Regex Filtering Categories

To distill the raw collection to 1,215 relevant posts, we applied regex pattern matching across three linguistic axes:

- **Action Patterns:** Explicit self-medication behaviors and access cues (e.g., `self[-]?medicat`, `self[-]?treat`, `self[-]?care`).
- **Substance Patterns:** Mentions of over-the-counter or unprescribed substances.
- **Target Conditions:** Ailments or symptoms the user was attempting to treat.

B Example Model Predictions

Table 7 presents representative predictions from the full pipeline, illustrating a true positive (row 1), false positives (rows 2–3), and a false negative (row 4).

C Error Taxonomy

Table 8 lists error categories derived from manual inspection of pipeline failures, particularly on out-of-domain IBM Debater text. Qualitative examples for three representative categories follow.

Narrative Framing.

> *"Wow this sounds like a female version of me. Early 20s I started with OTC then Rx codeine for back pain…"*

This instance expresses lived experience without making a generalizable assertion. The model predicts a claim due to narrative intensity and contrastive phrasing, but the gold label marks it as non-claim.

Advice vs. Claim.

> *"I'd definitely drink more water and try OTC migraine medicine."*

Advice lacks truth-conditional commitment, yet surface modality resembles normative claims, triggering a false positive.

Input Text	Gold	Pred	Conf.	Predicted Span
Just because an MRI shows a disc bulge doesn't mean it's the source of pain.	1	1	0.75	Just because an MRI shows a disc bulge doesn't mean it's the source of pain.
I'd definitely drink more water and try over the counter migraine medicine.	0	1	0.52	I'd definitely drink more water and try over the counter migraine medicine.
I don't remember the exact cocktail but I remember it was a bunch of Prozac.	0	1	0.60	I don't remember the exact cocktail but
Most of the warnings on OTC drugs are for liability protection.	1	0	0.43	

Table 7: Representative predictions from the claim detection and span extraction pipeline. Row 1: true positive. Rows 2–3: false positives. Row 4: false negative.

Error Type	Description
Narrative framing	Personal story without generalizable assertion
Experiential report	Lived experience only, no evaluative commitment
Implicit causality	Implied cause without truth-conditional commitment
Contrast-only	Opposition expressed without a claim
Advice vs. claim	Recommendation mistaken for assertion
Opinion-only	Subjective stance without truth claim
Question/clarification	Interrogative form
Factual statement	Descriptive but non-argumentative

Table 8: Error taxonomy derived from qualitative inspection of pipeline failures.

Experiential Report.

> *"I was doing the same as you, part prescription, part OTC… I was self medicating…"*

First-person experiential statements frequently trigger false positives, highlighting the difficulty of separating personal experience from evaluative assertion.

D Claim Typology Patterns

Table 9 lists the surface patterns used for rule-based claim typology assignment, as applied in Section 6.2.

E Hedging Marker Inventory

The following markers were used to identify epistemic hedging in claim spans. The list is not exhaustive but captures the most frequent constructions in Reddit health discourse: *I think*, *I feel*, *seems/appears*, *might/may/could*, *probably/possibly/likely*, *in my experience*.

F IBM Debater Dataset Fields

The cleaned IBM Debater test set contains the following fields per instance: `text` (sentence extracted from Wikipedia), `gold_claim` (binary claim label), and `query_type` (retrieval strategy label). Query types are: `q_strict` (strict evaluative claims), `q_that` (reported opinions), `q_mc` (mixed factual/argumentative), and `q_cl` (institutional or descriptive statements).

G Reproducibility

All experiments used fixed random seeds and deterministic 80:20 train/test splits. To reproduce results: install dependencies from `requirements.txt`, train models via scripts in `src/`, and run ablation and benchmark evaluations from `ablation_analysis/` and `benchmark_analysis/` respectively.

H Annotation Ambiguity Notes

Manual inspection revealed that annotators frequently relied on shared background knowledge, that implicit claims often hinge on conversational implicature, and that disagreements reflect discourse interpretation rather than labelling error.

The complete annotation guidelines including the primary claim rule, explicit vs. implicit priority rule, and worked examples drawn directly from the dataset are provided in `annotation_guidelines.pdf` in the project repository. These guidelines governed all labelling decisions and serve as the authoritative reference for reproducing the annotation scheme.

Type	Example Patterns
Causal	*because, leads to, results in*
Contrastive	*but, however, just because*
Normative	*should, need to, must*
Epistemic	*I think, seems, might, probably*

Table 9: Surface patterns used for rule-based claim typology assignment.